PIXAR • Lean Business Model Canvas

AGENTIC E-WASTE RECYCLING & COMPONENT HARVESTING PLATFORM

UN SDG 12: RESPONSIBLE CONSUMPTION | IBM SKILLSBUILD MASTERCLASS | TARGET 12.5 • CIRCULAR SILICON

PROBLEM 01

- **16.0 Lakh Tonnes** e-waste generated annually in India; 78% dumped into unorganized landfill streams.
- **Dormant Silicon Trap:** Millions of functional micro-motors, cameras, and chips sit idle in student and household drawers.
- **Toxic Soil Leaching:** Lead, Mercury, and Cadmium contaminate urban groundwater via crude acid-bath scrap handling.

EXISTING ALTERNATIVES

- *Local Kabadiwala:* No computer-vision appraisal, unsafe informal burning, zero component salvage.
- *Cashify / Trade-In:* Rejects older or semi-broken phones; zero maker lab distribution.
- *Passive Drop Bins:* Low public engagement, zero incentives, untracked logistics.

SOLUTION 04

- **Visual AI Scanner:** Instant ViT & YOLO model recognizes device, specs, chassis condition, and estimated residual value in seconds.
- **Component Harvesting Engine:** Automatically generates modular teardown blueprints for salvageable vibration motors, camera lenses, displays, and sensors.
- **Maker Lab Matching:** Real-time campus supply network routes harvested parts directly to university robotics & IoT projects.
- **Zero-Friction Doorstep Pickup:** Certified reverse-logistics partners collect unrepairable silicon.
- **Cryptographic EPR Seal:** Produces verifiable QR certificates calculating avoided carbon (CO_2e).

UNIQUE VALUE PROP 03

> *"Your Agentic AI Hardware Companion that Appraises, Salvages, Routes, and Certifies Dormant Electronics for a Second Life."*

- **For Citizens:** Instant fair value discovery & effortless doorstep eco-disposal.
- **For Student Makers:** 80% cheaper salvaged micro-components for robotics prototypes.
- **For Institutions:** Audited EPR certificates verifying zero-landfill compliance.

HIGH-LEVEL CONCEPT

"Cashify + HuggingFace + Blinkit + Cryptographic Circularity Protocol for Campus Electronics."

UNFAIR ADVANTAGE 09

- **Multi-Agent Visual Pipeline:** Proprietary dataset of Indian consumer electronics chassis & component teardowns.
- **Closed-Loop Campus Maker Demand:** Guaranteed local absorption of salvaged components eliminates warehousing overhead.
- **Verifiable Cryptographic Proof:** Tamper-proof QR verification hash compliant with India E-Waste Management Rules 2022.
- **Institutional Trust Network:** Direct integration with university engineering departments and IBM SkillsBuild student ecosystem.

CUSTOMER SEGMENTS 02

- **College Students & Households:** Holding 2-5 dormant devices needing safe disposal or residual cash.
- **Campus Robotics & IoT Labs:** Engineering clubs needing cheap sensors, motors, and batteries.
- **Educational Institutions:** Universities seeking NAAC sustainability credits and SDG compliance.
- **Authorized Recyclers:** CPCB-certified facilities seeking pre-sorted, clean e-waste streams.

EARLY ADOPTERS

- College Robotics & Maker Clubs.
- IBM SkillsBuild & Lenovo Leap learners.
- Campus Hardware Hackathon participants.

KEY METRICS 08

- **Appraisal Volume:** Devices scanned & appraised per month.
- **Component Recovery %:** Target >85% functional component salvage.
- **E-Waste Diverted:** Metric kilograms kept out of landfills.
- **Maker Cost Savings:** ₹ saved by student inventors using salvaged silicon.
- **EPR Certificates Issued:** Verified corporate & college compliance tokens.

CHANNELS 05

- **Acquisition:** University campus workshops, IBM SkillsBuild Masterclasses, IEEE student chapters, Green Campus Hackathons.
- **Distribution:** Progressive Web App (PWA), WhatsApp camera-scan chatbot, Campus smart deposit kiosks.
- **Logistics:** On-demand hyper-local couriers (Blinkit/Porter API) & CPCB recycler fleet.

COST STRUCTURE 07

- **Technology:** Multi-modal Vision API inference, vector embeddings database, Cloud hosting (AWS/GCP), WhatsApp Business API.
- **Logistics:** Reverse-logistics courier subsidies for doorstep collections.
- **Operations:** Component sorting & testing verification kits for campus maker labs.
- **Compliance & Legal:** CPCB recycler partner audits and EPR verification maintenance.

REVENUE STREAMS 06

- **Recycler Commission:** 12–18% marketplace fee from certified e-waste processors for bulk sorted feedstock.
- **Maker Subscriptions:** ₹199/mo student lab pass for unlimited subsidized component access.
- **B2B EPR Compliance Auditing:** Institutional certifications for universities and corporate CSR teams.
- **Refurbished Subassembly Sales:** Tested display panels and batteries sold to DIY repair stores.